THE INTERSECTION OF TWO SCHEME THEORETICALLY DENSE OPEN SUBSPACES IS SCHEME THEORETICALLY DENSE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a two-step injectivity argument; the density characterization is used both as hypothesis and as conclusion.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Scheme theoretic density characterized by injectivity of the pushforward of the structure sheaf. Lemma lemma-characterize-scheme-theoretically-dense (3526-3532), used at proof lines 3562-3565. Is the criterion applied twice, once for each of U and V, and once for the conclusion.
- Etale localization of the density condition. used at proof lines 3559-3561. Reduces the claim to the displayed chain of restriction maps.
- Injectivity is preserved by composition. used at proof lines 3562-3563. Closes the two-step argument.

Source reference. Original lemma 0836: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10276–10276. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. If U, V are scheme theoretically dense open subspaces of X, then so is $U \cap V$.*

Proof. Let $W \to X$ be any étale morphism. Consider the map $\mathcal{O}(W) \to \mathcal{O}(W \times_X V) \to \mathcal{O}(W \times_X (V \cap U))$. By Lemma 0836 both maps are injective. Hence the composite is injective. Hence by Lemma 0836 $U \cap V$ is scheme theoretically dense in X. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.